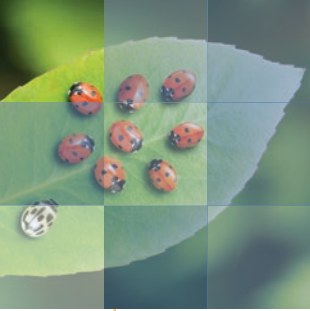




Binomial Distributions

A table or graph can be used to represent outcomes of a variable that has a binomial distribution.

Determining Whether a Variable Has a Binomial Distribution

A doctor knows that for a certain type of adult migraine headache, Medicine Q works about 70% of the time. If the medicine is tried with two or more different patients who have this condition, the resulting probability distribution is a binomial distribution.

Binomial Distribution

A variable has a binomial distribution if it has these traits:

- There are a fixed number of independent trials.
- The outcome of each trial is success or failure.
- The probability of success for each trial is the same.

This situation meets all of the conditions for a binomial distribution.

- There are a fixed number of independent trials; there are two trials, and each patient taking Medicine Q is independent of the other.
- The outcome of each trial is a success or failure; for each patient, the medicine will either help the migraine (success) or it will not (failure).
- The probability of success for each trial is the same; there is a 0.7 probability of success when each patient takes Medicine Q.

► **Think About It** An experiment with two or more coin tosses has a binomial distribution. The probability of success and failure is the same, 50%.

Creating Binomial Distributions

Suppose the doctor is seeing two patients who have this type of adult migraine headache. You can create a binomial distribution table or graph to represent outcomes for these two patients.

The variable X can take on three different values: Medicine Q will work for none of the patients, one of the patients, or both patients. So the variable X is equal to 0, 1, or 2.

► **Remember** If A and B are independent events, then

$$P(A \text{ and } B) = P(A) \cdot P(B).$$

When $X = 0$, it means that Medicine Q failed to work for both patients. Since $P(\text{success}) = 0.7$, $P(\text{failure}) = 0.3$.

$$\begin{aligned} P(X = 0) &= P(\text{failure}) \cdot P(\text{failure}) \\ &= 0.3 \cdot 0.3 \\ &= 0.09 \end{aligned}$$

When $X = 1$, it means that the medicine worked for one of the patients but not both.

$$\begin{aligned} P(X = 1) &= P(\text{success}) \cdot P(\text{failure}) + P(\text{failure}) \cdot P(\text{success}) \\ &= 0.7 \cdot 0.3 + 0.3 \cdot 0.7 \\ &= 0.21 + 0.21 \\ &= 0.42 \end{aligned}$$

When $X = 2$, it means that the medicine worked for both patients.

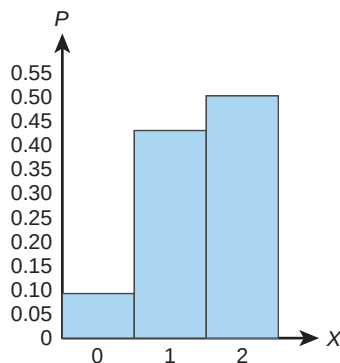
$$\begin{aligned}P(X = 0) &= P(\text{success}) \cdot P(\text{success}) \\&= 0.7 \cdot 0.7 \\&= 0.49\end{aligned}$$

You can now use the probabilities that you calculated for each outcome to create a binomial distribution table and a binomial distribution histogram.

Binomial Distribution Table

X	0	1	2
P	0.09	0.42	0.49

Binomial Distribution Histogram



Interpreting Binomial Distributions

The doctor would be happy if the medicine was successful with at least one of the two patients. In the language of probability, the question becomes what is $P(X \geq 1)$?

The answer to this question is found by adding the appropriate probabilities in the table.

$$\begin{aligned}P(X \geq 1) &= P(X = 1) + P(X = 2) \\&= 0.42 + 0.49 \\&= 0.91\end{aligned}$$

There is a 91% probability that Medicine Q will be successful with at least one of the patients.